

Blaise Baptist

Devon, PA | baptisbc@rose-hulman.edu | linkedin.com/in/blaise-baptist | github.com/BlaiseBaptist

Computer Science student seeking a software internship for Summer 2027

EDUCATION

Rose-Hulman Institute of Technology

Terre Haute, IN

Bachelor of Science in Computer Science

Expected May 2029

Coursework: Operating Systems, Computer Architecture, Data Structures & Algorithm Analysis, Systems Programming, Object-Oriented Software Development, Digital Systems; *in progress*: Web Programming, Debugging

EXPERIENCE

Thornbots, Rose-Hulman RoboMaster Team

Sept 2025 – Present

Secretary; Software Developer, Sentry Robot

Terre Haute, IN

- Built the ROS 2 localization stack for the autonomous Sentry robot, with SLAM, AMCL, and EKF backends that fuse wheel odometry with rf2o laser odometry
- Wrote the targeting stage of the CV pipeline, which tracks enemy robots from detected armor panels and decides where to aim and when to fire; co-wrote the depth step that turns 2-D detections into 3-D panels
- Wrote Gazebo test benches that measure localization drift against ground truth and score aiming and tracking
- Ported the full robot stack from ROS 2 Humble to Jazzy and Isaac ROS 4.6

Association Resource Center

July – August 2025

Coding Intern

Milwaukee, WI

- Wrote a Rust tool that tags articles automatically, replacing a manual step for staff
- Worked with a team to build a front-end website in Rust, HTML, and CSS

PROJECTS

MediaManager | Python, Django, Rust, ffmpeg

2026

- Built a client-server system that splits AV1 encoding across 5 machines, saving 1.5 TB on a 6 TB media library
- The Django server locks each job to one worker, requeues jobs from workers that stall, and skips files already in AV1
- The Rust client runs parallel jobs on the machine's hardware AV1 encoder, or on SVT-AV1 if it has none

Mobile Home Lab | Ubuntu, Docker, Tailscale, iptables

Aug 2025 – Present

- Set up a Protectli Vault as a portable router that shares an Ethernet or LTE uplink over its own Wi-Fi and works as a Tailscale exit node
- Rewired it to run 21 hours over USB-C on custom Li-ion cells; it also hosts Pi-hole and Vaultwarden

Newton Fractals | Rust

2024 – 2025

- Wrote a multithreaded renderer for Newton's-method fractals that has produced 10-gigapixel images and includes an interactive viewer

RS-232 Serial Port Reader | Rust, Iced

Sept 2024 – May 2025

- Built a desktop app that splits one serial port's Arduino sensor stream into live graphs and saves it as CSV

VEX Robotics | Python, Fusion 360

Aug 2021 – May 2025

- Designed robots in Fusion 360 and wrote a PID controller for autonomous driving; the team qualified for State three times and Worlds once

SKILLS

Languages Rust, C++, C, Python, Java, HTML/CSS/JavaScript

Robotics ROS 2 (Humble, Jazzy), Isaac ROS, NVIDIA Jetson, Gazebo, SLAM/AMCL/EKF, YOLOv8, RealSense

Tools Linux administration, Docker, Git/GitHub, Django, ffmpeg, Vim, Fusion 360

ACTIVITIES

Lifeguard, Nassau Swim Club, 2024 (CPR/AED certified) | **Theatre Tech**, The Grayson School, 2024–2025 (sound board)